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Scaling reliable interatomic potentials to complex nuclear alloys via pretrained atomic models

Mingxuan Jiang^{1,5}, Biao Xu^{1,2,5}✉, Yixin Deng³, Shihua Ma³, Ji-Jung Kai²✉, Fei Gao⁴ & Huiqiu Deng¹✉

Next-generation fission and fusion reactors impose unprecedented demands on structural materials, requiring simultaneous resistance to high temperatures, high-dose irradiation, and aggressive corrosion. Designing materials that harness the intrinsic properties of multiple elements and their synergistic interactions has emerged as a key strategy to achieve such integrated performance. To guide this design paradigm, a mechanistic understanding of chemically and structurally complex systems is essential. However, such understanding is currently constrained by the lack of high-fidelity interatomic potentials (IAPs) that enable predictive, large-scale atomistic simulations. Here, we employ, for the first time, a multi-task, physics-informed pretraining strategy with the large atomic model (LAM) to systematically evaluate the construction and predictive capability of IAPs for complex nuclear alloy systems. Using Ta-Nb-W-Mo-V as a representative case, the resulting DPA2-5E model—trained solely on the quinary dataset—significantly outperforms conventional machine learning IAPs, demonstrates superior transferability to lower-order subsystems, and accurately reproduces cascade damage and stress-strain behavior. Furthermore, this approach extends to nuclear-relevant structures and corrosive/oxide environments, enabling high-fidelity IAPs and large-scale simulations at reactor extremes.

Next-generation fission nuclear reactors and future fusion reactors will confront materials with more intense radiation fields, higher operating temperatures, aggressive corrosive coolants, and severe mechanical loads^{1,2}. These harsh service conditions push conventional structure materials (e.g., austenitic steels and zirconium alloys) beyond their performance limits and present a great challenge to the development of future reactor materials^{3–5}. One promising response is to design the materials that intentionally combine multiple principal elements to exploit both the intrinsic properties of each constituent and their synergistic interactions⁶. Among various candidates, multi-principal element alloys (MPEAs), including high-entropy alloys (HEAs), have emerged as compelling options for advanced nuclear systems⁷. Through compositional tuning, Al or Ti can reduce density and refine microstructure^{8,9}, improving strength and toughness; W, Ta, and Nb (high melting points) enhance high-temperature stability^{10,11}. The complex mixing of elements can also produce overall properties superior to those of any traditional alloys, as exemplified by the four core effects observed in HEAs (high-entropy effect, lattice distortion, sluggish diffusion, and cocktail

effect)^{12–14}. These effects may collectively suppress defect mobility and enhance defect recombination, thereby improving irradiation resistance^{15–17}. In addition, Tailored microstructures further amplify these benefits. For instance, nanoscale L1₂-Ni₃Al precipitates in face-centered cubic (FCC) matrices can raise strength without sacrificing ductility^{18,19}. In short, compositional and structural complexity offers a pathway toward meeting the combined performance demands of nuclear environments.

To translate this promise into structural materials suitable for advanced nuclear reactors, it is essential to understand the underlying microscopic mechanisms. For certain critical components—such as reactor cores, claddings, or structural supports—service lifetimes may extend over several decades^{3–5}. In these cases, the microstructural evolution under extreme conditions of high temperature, intense irradiation, and corrosive environments becomes particularly important for predicting long-term performance and reliability. Experimentally, probing the dynamic atomic-scale processes—such as cascade damage evolution—in these complex alloys remains highly challenging due to limitations in controlling

¹School of Physics and Electronics, Hunan University, Changsha, China. ²Center for Advanced Nuclear Safety and Sustainable Development, City University of Hong Kong, Hong Kong, China. ³School of Materials Science and Engineering, Hunan University, Changsha, China. ⁴Department of Nuclear Engineering and Radiological Science, University of Michigan, Ann Arbor, MI, USA. ⁵These authors contributed equally: Mingxuan Jiang, Biao Xu.

✉ e-mail: xubiao1189@hnu.edu.cn; jijkai@cityu.edu.hk; hqdeng@hnu.edu.cn

irradiation dose and temperature, restricted spatial and temporal resolution, and the inherent difficulties of capturing non-equilibrium processes in realistic or harsh physical environments^{20–22}. Given these experimental limitations, molecular dynamics (MD) simulations serve as a key approach to access atomic-scale insights, especially under extreme or transient conditions^{23,24}. Ab initio molecular dynamics simulation based on density functional theory (DFT) can provide accurate atomic-scale insights, but its applicability to nuclear materials is severely limited by its high computational cost, which restricts simulations to hundreds of atoms and picosecond timescales—far below the spatial and temporal scales required to capture defect evolution and damage accumulation in complex MPEAs systems^{25,26}. In contrast, classical molecular dynamics using empirical interatomic potentials (IAPs) (e.g., as implemented in LAMMPS²⁷) can simulate systems with millions of atoms over nanosecond to microsecond timescales, enabling the study of phenomena such as primary radiation damage events^{25,26,28,29}. However, developing accurate empirical IAPs for complex MPEAs remains extremely challenging due to the vast configurational space, complex multi-element interactions, and the limited transferability of traditional potential forms³⁰.

In recent years, the emerging machine-learning IAPs (MLIAPs), such as Gaussian Approximation Potential (GAP)³¹, Deep Potential (DP)³², and Deep Potential with Attention (DPA1)³³, offer significantly improved accuracy over empirical IAPs due to their flexible functional forms, ability to capture complex many-body interactions, especially in complex systems. While MLIAPs offer a theoretically viable solution to modeling such complex systems, their practical deployment is fundamentally constrained by the exponential growth of configurational space with increasing compositional complexity. To ensure robustness, transferability, and physical fidelity, conventional MLIAPs must be trained from scratch on both the target composition alloy and a wide range of its lower-order subsystems. This necessitates a massive and diverse training dataset on each system and lower-order subsystem. Since each data point typically relies on DFT calculations—which are computationally intensive—the total cost quickly becomes prohibitive. For example, even with advanced active-learning techniques, constructing a high-fidelity full composition MLIAPs for a ternary alloy (Al-Mg-Cu) required 141K first-principles labeled data, costing more than 10 million CPU-hours³⁴. Consequently, most existing MLIAPs remain limited to unary or binary systems³⁵, while high-quality potentials for more complex MPEAs remain rare. This has left the mechanistic modeling of complex nuclear materials largely unexplored.

Pretrained large atomic models (LAMs) offer a path to lower labeling cost by transferring physics-aware representations learned from broad Quantum chemical datasets to complex target systems. Domain-specific LAMs have appeared across materials and molecular domains—for example, models trained on the MPtrj inorganic trajectories (e.g., MACE-MP-0, SevenNet-0) and models for small molecules (e.g., AIMNet, Nutmeg) at well-defined quantum-chemistry levels of theory^{36–39}. These models typically excel in-distribution, yet their generalization to multicomponent, reactor-relevant complex MPEAs is uneven, in part because pretraining corpora usually under-represent high-order composition spaces, defect-rich non-equilibrium states, and elevated-temperature configurations central to nuclear applications. Another major limitation is that most existing datasets are drawn from a single domain, whereas large amounts of quantum-mechanical data have also been accumulated in other fields but remain difficult to integrate for pretraining, owing to inconsistencies in computational settings such as exchange–correlation functionals, pseudopotentials, and codes used in DFT calculations. Recent multi-task pretraining strategy (called DPA-type) addresses data heterogeneity by jointly learning from DFT datasets produced with different exchange–correlation functionals, basis sets, and codes, and they empirically obey scaling laws whereby capability grows with model size, data volume, and computation resources^{33,40,41}. Nevertheless, systematic application of such multi-task LAMs to modeling nuclear complex material systems remains limited.

To close this gap, we systematically compare a multi-task pretrained LAM (DPA2) against conventional from-scratch pipelines for complex

nuclear materials. This work represents the first application of multi-task pretraining to nuclear structural alloys, providing critical insights into its ability to enhance generalization across diverse chemistries and defect states that are central to reactor performance. Using Ta-Nb-W-Mo-V as a representative case, we fine-tune DPA2 using only a small, highest-component dataset from the target quinary alloy system (no labels from lower-order subsystems). Under matched data budgets, the fine-tuned model achieves higher in-domain accuracy on quinary configurations and exhibits strong downward transfer to quaternary/ternary/binary/unary subsets. In addition, it reproduces key signatures of irradiation-damage evolution and temperature-dependent mechanical response with fidelity comparable to a control trained on an “All” dataset (~30 subsystems), underscoring its promise for applications that target nuclear-material-specific behaviors. Importantly, in order to show its broad applications, the same recipe extends beyond compositional complexity to structural complexity: it remains accurate for coherent L1₂-precipitate-strengthened FCC microstructures and for interface-dominated liquid-Pb-Bi environments with and without oxygen, with consistently small train–validation gaps. This establishes a practical and scalable route to reliable MLIAPs for complex nuclear materials simulation without re-collecting exhaustive subsystem datasets.

Results

The performance of LAM-based DPA2-MLIAPs

The performance of MLIAPs for complex MPEAs is fundamentally influenced by two key factors: the quality of the training dataset and the learning strategy used to construct the MLIAPs^{42,43}. However, to meaningfully improve MLIAPs in practical applications, it is important to understand the individual contributions of these two aspects. In particular, reducing the dependency on expensive and extensive labeled datasets is crucial for making MLIAPs development accessible to a wider range of researchers working on advanced nuclear materials design or optimization. To this end, instead of creating a bespoke dataset that may inadvertently favor our model (e.g., LAM-based DPA2), we chose a publicly available dataset previously used to train the tabGAP potential for the Ta-Nb-W-Mo-V system⁴⁴. This dataset has a manageable size, representing a realistic data budget for most academic or industrial projects, and serves as a fair, unbiased benchmark for comparing training strategies in this study. Importantly, this dataset also exhibits high diversity, which is critical for testing the generalization and transferability ability of MLIAPs to unseen configurations within the same system and their transferability to related subsystem alloys. It includes the quinary Ta-Nb-W-Mo-V alloy system (a relatively small dataset containing only several hundred configurations), and all its 30 unary, binary, ternary, and quaternary subsystems ($5 + 10 + 10 + 5 = 30$), for a total of 31 alloy systems. Within each system, a wide range of atomic environments are present—including crystalline structures with vacancies and interstitials, surface slabs, liquids, chemically ordered phases, and transition-state geometries—ensuring comprehensive configurational coverage (Fig. 1). Detailed information on the dataset is provided in Supplementary Materials S1.

With this fixed and representative dataset, we systematically compared the performance of our LAM-based DPA2-MLIAPs—fine-tuned from a multi-task pretrained LAM DPA2_{medium_28_10M_rc0.pt} (www.aissquare.com/models/)—with a conventional MLIAP DPA1 trained from scratch. By doing so, we aimed to decouple the contributions of (1) the training strategy—multi-task pretrained LAM fine-tuning versus conventional training from scratch—and (2) the scope of the training dataset—quinary-only versus full multicomponent coverage. The dataset pre-processing and model training workflows for both DPA2 and DPA1 approaches are illustrated in Fig. 2, with detailed methodological descriptions provided in the “Methods” section. This setup allows for a direct comparison of performance across models trained under different learning paradigms and dataset coverage levels.

To systematically assess the impact of model architecture and dataset scope, we constructed and benchmarked five representative MLIAPs. As

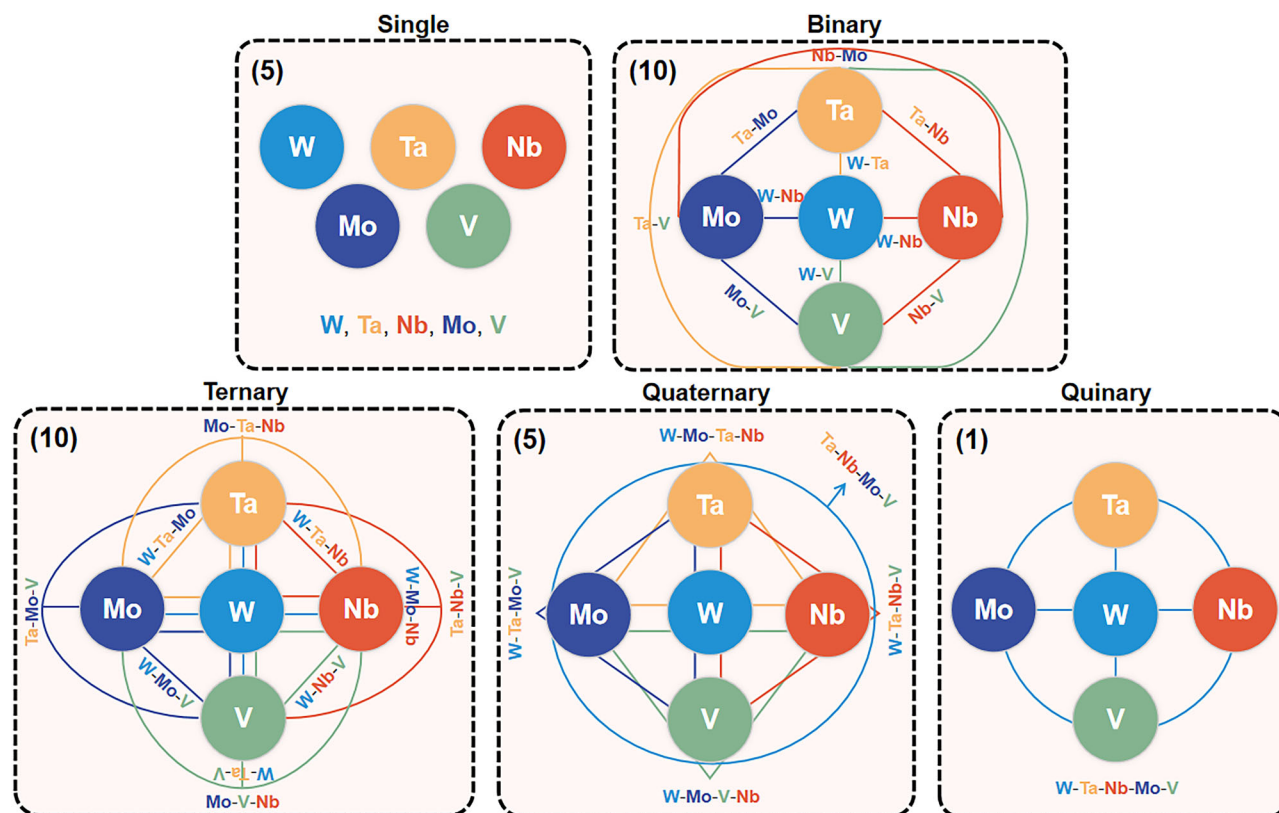


Fig. 1 | Schematic of the diverse training dataset encompassing the Ta-Nb-W-Mo-V quinary alloy and all its subsystems. Colored circles represent the five constituent elements (W, Ta, Nb, Mo, and V). In addition, the colored lines and closed loops connecting or enclosing multiple circles denote alloy subsystems

formed by the corresponding combination of elements. **a** Five unary (single element) systems, **b** 10 binary combinations, **c** 10 ternary combinations, **d** five quaternary combinations, and **e** a quinary system, totaling 31 alloy systems in the dataset.

Fig. 3 shows, these include: DPA2-All, DPA1-All, and a reference model tabGAP-All (a tabulated GAP model from a recent study^{42,44})—with all three trained on the full, diverse dataset (referred to as Dataset-All), which comprises 31 distinct alloy systems, including Ta-Nb-W-Mo-V and all its unary, binary, ternary, and quaternary subsystems. Additionally, two MLIAPs—DPA2-5E and DPA1-5E—were trained using only the quinary subset of this dataset. The global root-mean-square error (RMSE) for energy and force predictions is summarized in Fig. 3, in which the RMSE of the tabGAP-All was directly obtained from literature^{42,44}.

Among the models trained on Dataset-All, DPA2-All exhibits the lowest RMSE for both energies and forces on the overall training and validation sets, as shown in Fig. 3a–d. In particular, on the validation dataset performance, the DPA2-all achieves significantly better energy and force prediction accuracy on each composition dataset, compared to the literature-reported tabGAP-All and DPA1-All models. In the small-data scenario, DPA2-5E also outperforms DPA1-5E on the validation datasets, yielding lower RMSE for both energy and force in Fig. 3c, d. These findings demonstrate that under the same dataset condition, the LAM-based MLIAPs consistently deliver significantly superior accuracy and generalization. We attribute such superior performance to the effective knowledge transfer enabled by the multi-task pretrained LAM, which provides a physically informed prior and enhances model robustness.

Beyond generalization to unseen configurations within the same system, we further assess the transferability of MLIAPs—specifically, whether a model trained solely on a complex alloy can accurately predict energies and forces for its simpler subsystems. To investigate this, we compare DPA2-5E and DPA1-5E (trained only on the complex system) with DPA2-All and DPA1-All (trained on the full 31-system dataset), separately. These comparisons disentangle the respective roles of model architecture and dataset composition in determining transfer performance.

First, we observe that DPA2-5E transfers remarkably well to unseen subsystems: its energy and force errors on quaternary and ternary subsets remain close to those of DPA2-All, which suggests that the models trained solely on the quinary dataset can accurately predict energies and forces for the quaternary and ternary subsets in such a complex system. However, such transability cannot belong only to the model architecture; the dataset also contributed. In Fig. 3c, d, a similar trend appears in the DPA1 model family: DPA1-5E also shows reasonable transferability to its subsystems, although with larger performance degradation compared to DPA2-5E, which means the intrinsic structural and compositional link between the complex alloy and its subsystems also play a nonignorable role in such great transability. However, the LAM-based DPA2 model DPA2-5E shows consistently higher accuracy and more stable transfer performance than DPA1-5E under the same data scope. This highlights that while the inherent structure of the dataset enables a degree of transferability, the use of pre-trained model priors further enhances the ability to extract and generalize physical patterns from limited data, especially in chemically diverse and structurally complex alloy systems.

Beyond its strong transferability, DPA2-5E also departs from conventional data-hungry training paradigms. Traditionally, achieving high accuracy for complex alloys has required comprehensive datasets covering all relevant subsystems, as in DPA1-All, which is trained from scratch on the full 31-system dataset. By contrast, DPA2-5E, trained only on quinary data, delivers even higher accuracy across the quinary, quaternary, and ternary subsets (Fig. 3c, d). This improvement stems from the physical knowledge embedded in the pretrained LAM, which allows DPA2-5E to efficiently capture key atomic interaction patterns and avoid redundant relearning. These results demonstrate that LAM-based multi-task pretraining can drastically reduce data-labeling costs while maintaining or even improving model fidelity, opening new avenues for mechanistic atomistic simulations

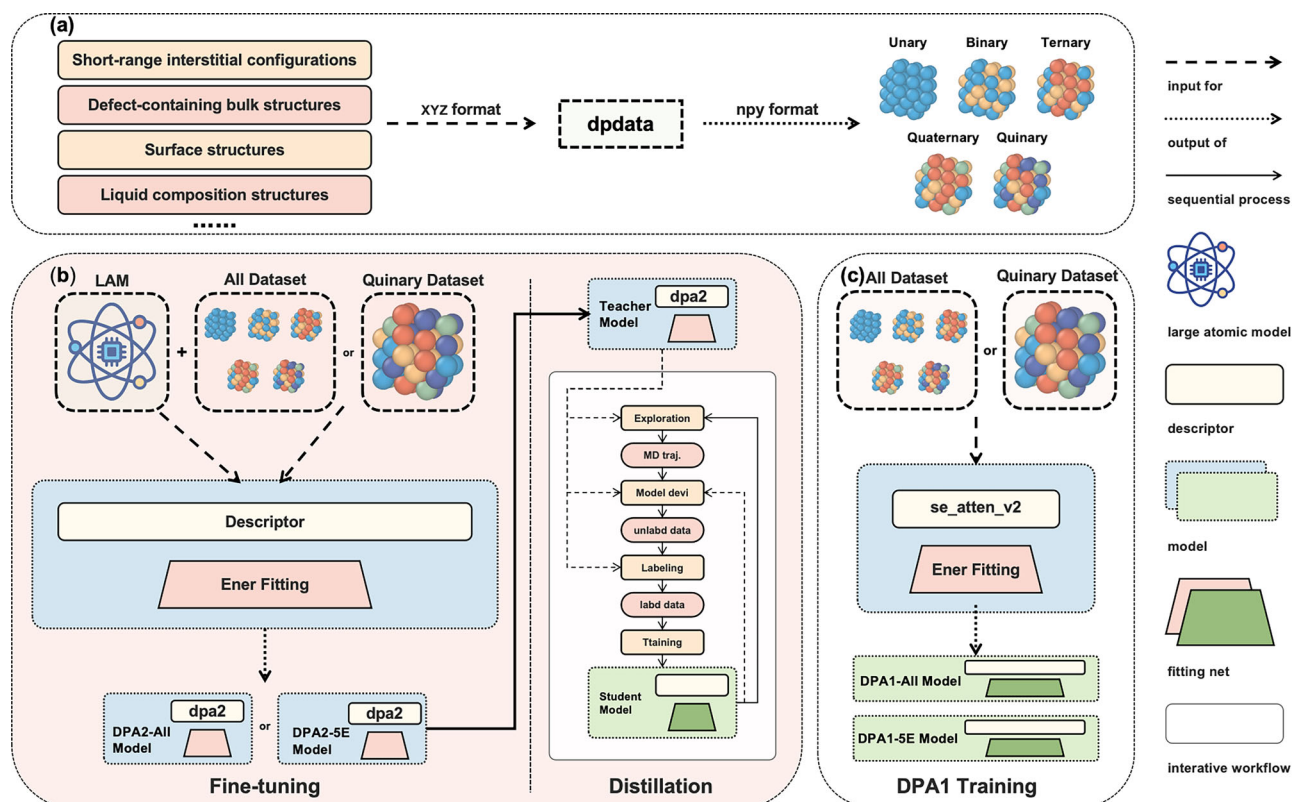


Fig. 2 | Schematic illustration of the model training workflows. **a** Conversion of the dataset format. **b** Fine-tuning and distillation procedure for the LAM-based DPA2 model. **c** Training process for the DPA1 model.

of complex nuclear materials that have long been constrained by data scarcity and high compositional complexity.

Cascade simulation

In the section “The Performance of LAM-based DPA2-MLIAPs” above, although trained on a smaller dataset, the DPA2-5E potential demonstrates fitting errors close to those of the baseline MLIAP DPA2-All, suggesting comparable accuracy. However, subtle differences between the two MLIAPs still exist. We next investigate whether such a difference in MLIAPs affects the predicted radiation damage behavior in the MPEAs. In particular, we quantify how well an MLIAP built with only complex-alloy data can reproduce the results of a full-dataset MLIAP in cascade simulations relevant to nuclear materials. We performed a series of displacement cascade simulations in three representative alloy systems: the quinary Ta-Nb-W-Mo-V alloy, a representative quaternary subset (Ta-Nb-W-Mo), and a binary subset (Ta-W). We simulated cascades at 300 K with three primary knock-on atom (PKA) energies (2 keV, 5 keV, 10 keV) using both the DPA2-5E and DPA2-All potentials (with the latter serving as a high-accuracy reference). The detailed settings are provided in the “Methods” section “Simulation methods.” Figure 4 compares the number of primary Frenkel pairs (FPs) defects surviving at the peak of the thermal spike and after cooling to 300 K equilibrium for the two MLIAPs, as a function of PKA energy. The error bars denote the standard deviation of the number of FPs measured at both the thermal-spike and post-relaxation stages, computed over 20 independent cascade runs with different random seeds/ knock-on directions. We find that, in Fig. 4a, the DPA2-5E and DPA2-All potentials predict virtually identical numbers of FPs at both the peak and after thermal relaxation, across all cascade energies. This indicates that the complex-only MLIAP DPA2-5E can accurately reproduce radiation damage evolution in the target quinary alloy.

To evaluate transferability performance in the radiation context, we also ran cascades in the 4-element Ta-Nb-W-Mo alloy and the 2-element Ta-W alloy using the DPA2-5E potential, and compared the outcomes to

those from DPA2-All. The results (Fig. 4b, c) show that the DPA2-5E potential again yields defect production levels that closely match the reference model in both the quaternary and binary systems. For instance, the difference in surviving FPs count between DPA2-5E and DPA2-All is negligible (within statistical variation) for all PKA energies tested. These findings demonstrate that the DPA2-5E potential accurately captures radiation-induced defect behavior even in lower-order subsystems not included in training, highlighting its practical transferability for radiation damage simulations in complex alloy systems.

Furthermore, we analyzed the composition of the FPs produced, focusing on the interstitial atoms after a 10 keV cascade (Fig. 4d–f). The elemental fractions of interstitials (and likewise of vacancies, not shown) predicted by DPA2-5E are virtually indistinguishable from those predicted by DPA2-All in each alloy. In other words, not only the total number of defects but also the defect chemistry is captured consistently by the complex-only potential. This capability to accurately capture both the quantity and elemental identity of radiation-induced defects makes the LAM-based MLIAPs especially valuable for modeling irradiation mechanisms in complex alloys, enabling insights into element-specific defect dynamics and long-term damage evolution.

In summary, despite being trained solely on a complex quinary alloy system, the DPA2-5E model reliably captures radiation-induced defect generation and evolution—not only in the target quinary system but also in related lower-order alloys. Its ability to reproduce both quantitative and compositional aspects of defect populations demonstrates that LAM-based MLIAPs can reliably capture fundamental irradiation mechanisms. This transferability underscores the model’s utility for efficient and mechanistically insightful simulations of complex nuclear materials, especially under conditions where full compositional data is impractical to obtain.

In summary, although trained only on the complex quinary alloy, DPA2-5E reliably captures radiation-induced defect production and evolution in both the quinary and related lower-order alloys. Its ability to reproduce the quantitative and compositional features of defect populations

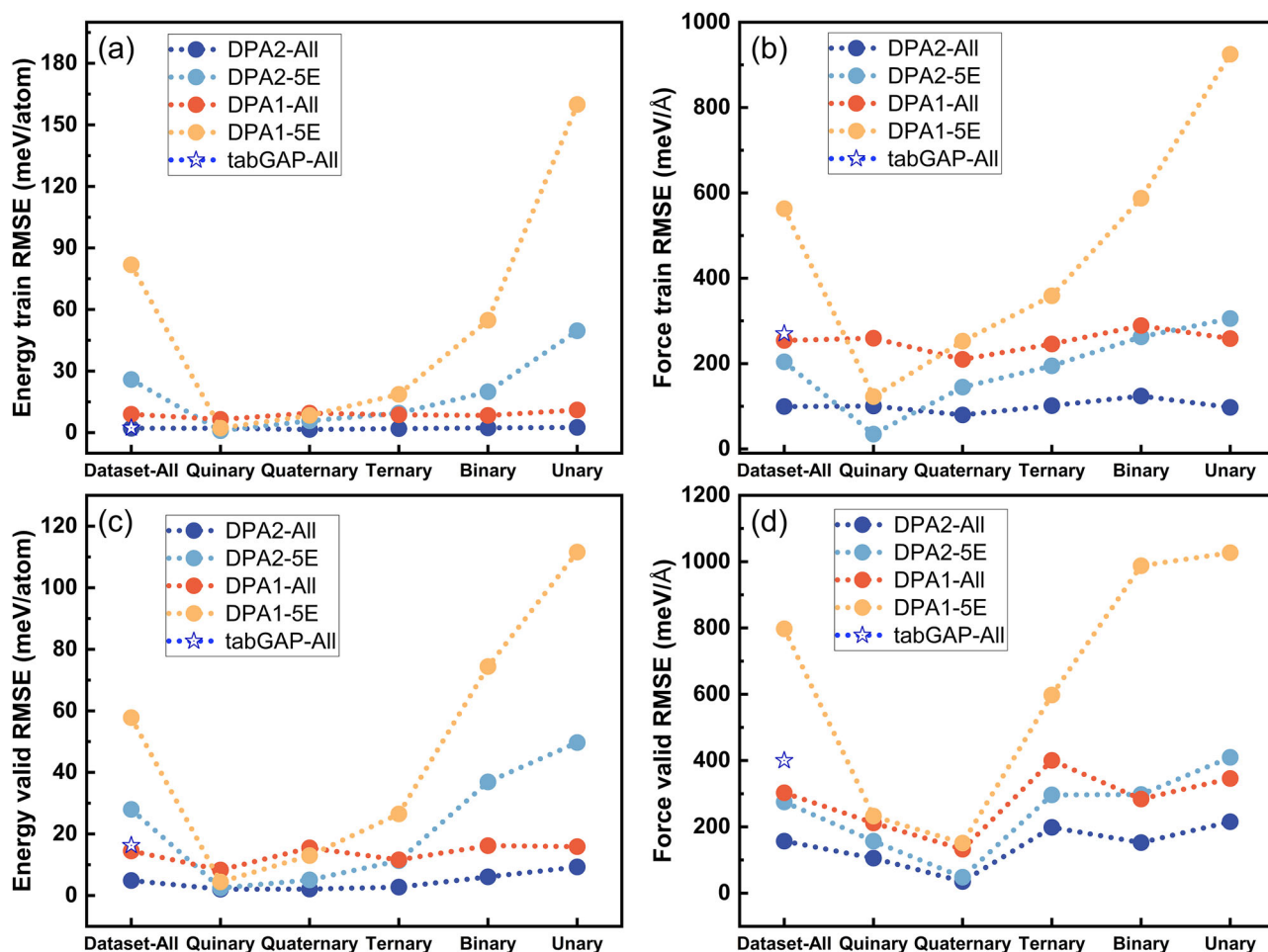


Fig. 3 | Energy and force prediction performance for five MLIAPs generated from different datasets (Dataset-All includes the quinary, quaternary to unary datasets) and training methods. a, b The train performance of five MLIAPs on different datasets. **c, d** The valid performance of five MLIAPs on different datasets. Dark-blue

and light-blue filled circles denote the DPA2-All and DPA2-5E models, respectively; dark-orange and light-orange filled circles denote the DPA1-All and DPA1-5E models, respectively; the open blue star represents the tabGAP-All reference potential. The lines connecting the symbols are guided to the eye.

shows that LAM-based MLIAPs can capture key irradiation mechanisms and are well-suited for efficient, mechanistically insightful simulations of complex nuclear materials, especially when full compositional datasets are impractical.

Mechanical properties simulation

Beyond radiation damage, mechanical strength is a critical design parameter for structural materials in nuclear applications. To assess whether the MLIAPs trained on limited complex-alloy data can reliably predict mechanical performance, we evaluated the uniaxial compressive yield strength of the quinary Ta-Nb-W-Mo-V alloy and its two subsystems (Ta-Nb-W-Mo and Ta-W), using both the DPA2-5E and DPA2-All potentials across a wide temperature range (0–2100 K).

As shown in Fig. 5a, the DPA2-5E model—despite being trained solely on the complex quinary alloy—accurately captures the temperature-dependent trend of yield strength in the Ta-Nb-W-Mo-V system, closely matching the full-data reference model DPA2-All across the entire temperature range. This result demonstrates that a multi-task pretrained LAM-based MLIAPs can reliably predict temperature-dependent mechanical behavior in target complex alloys, even under small-data constraints. Notably, in Fig. 5, error bars denote standard deviation computed over $n = 10$ independently generated random atomic configurations per composition/condition; each configuration was relaxed and loaded under identical protocols, and plotted symbols indicate the mean value.

When extended to compositionally simpler subsystems, Fig. 5b, c shows that DPA2-5E continues to faithfully reproduce the temperature-dependent trend of yield strength in both the quaternary (Ta-Nb-W-Mo) and binary (Ta-W) alloys. This result highlights the model's strong transferability and robustness in capturing key mechanical behaviors across varying compositional complexities—even for systems not included in the training data.

Taken together, these results establish LAM-based MLIAPs as robust, generalizable, and data-efficient tools for predicting mechanical performance, paving the way for inverse design and high-throughput screening of advanced alloys for nuclear and other extreme environments. At the same time, DPA2-5E systematically underestimates yield strength, especially in chemically simpler subsystems; the origin of this deviation and possible remedies are examined in the Discussion.

Discussion

The use of multiple principal elements aims to exploit distinct elemental properties and synergistic effects, offering a promising route to materials that can withstand high temperatures, intense irradiation, and corrosive environments in advanced nuclear systems. However, as compositional complexity increases, constructing reliable MLIAPs becomes increasingly challenging: the configurational space grows combinatorially, ab initio data over an exponentially large compositional space are prohibitively expensive, and it is difficult to balance accuracy and computational cost. Our results

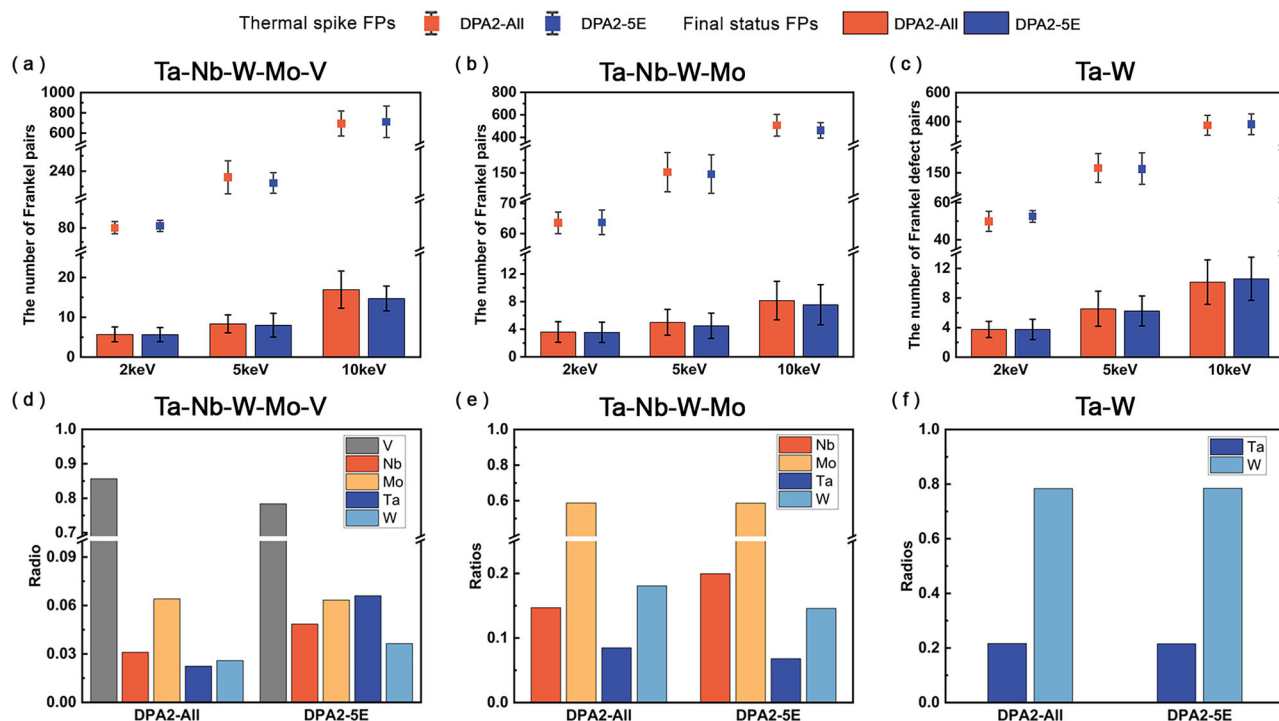


Fig. 4 | Primary radiation damage outcomes predicted by the DPA2-All and complex-only DPA2-5E. Number of Frenkel pairs produced at the peak of the cascade (scatter) and after thermal equilibration at 300 K (bars) as a function of primary knock-on atom energy, for (a) the quinary Ta-Nb-W-Mo-V alloy, (b) the Ta-Nb-W-Mo alloy, and (c) the Ta-W binary alloy. Red and blue symbols/bars correspond to the DPA2-All and DPA2-5E models, respectively. The error bars are derived from the standard deviation of FPs at the thermal spike state and after

relaxation across 20 separate simulations. Elemental composition of interstitial defect atoms in the equilibrium state after a 10 keV cascade, for (d) Ta-Nb-W-Mo-V, (e) Ta-Nb-W-Mo, and (f) Ta-W. The different colors within each stacked bar distinguish the constituent elements (V, Nb, Mo, Ta, and W) as indicated by the legends, and the height of each colored segment gives the fraction of interstitials associated with that element.

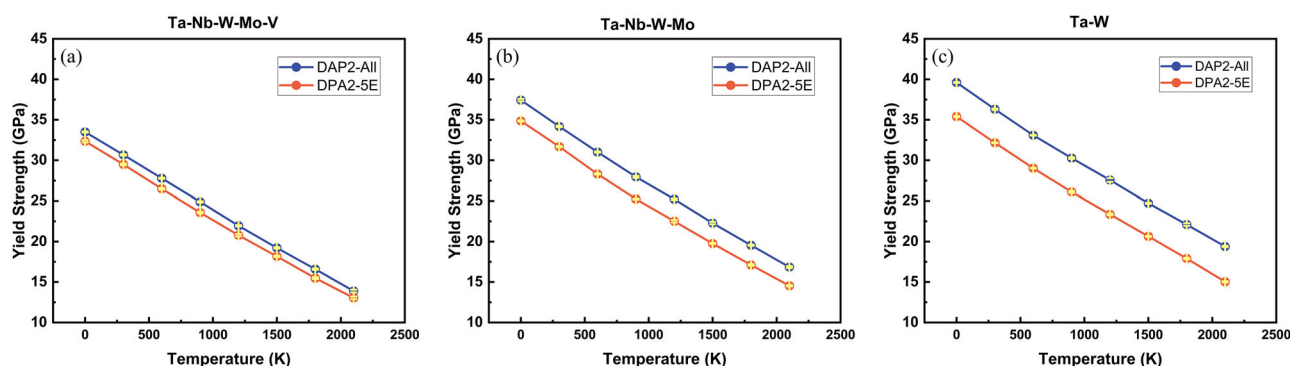


Fig. 5 | Temperature dependence of compressive yield strength for the complex alloy system and its subsystems predicted by LAM-based MLIAPs. Variation of compressive yield strength with temperature for the equimolar Ta-Nb-W-Mo-V, Ta-Nb-W-Mo, Ta-W alloy, as predicted by the LAM-based DPA2-5E and the

reference DPA2-All a Ta-Nb-W-Mo-V, b Ta-Nb-W-Mo, and c Ta-W. Blue and orange lines correspond to the DPA2-All and DPA2-5E models, respectively. The yellow symbols with vertical error bars indicate the mean values and one standard deviation across the 10 realizations.

show that a multi-task pretrained LAM provides a practical way through this bottleneck. By fine-tuning the pretrained model on a modest amount of Ta-Nb-W-Mo-V data, we obtain an MLIAP that reaches near-DFT accuracy in energies and forces, outperforms both a model trained from scratch and available MLIAP potentials, and enables large-scale simulations of radiation damage and mechanical behavior in a complex nuclear structural alloy.

A second key outcome is that the fine-tuned DPA2-5E model, although trained only on the quinary alloy, transfers well to many related quaternary, ternary, and binary subsystems. This suggests a practical strategy for complex MPEAs: instead of training a separate MLIAP for every candidate composition, one can fine-tune and distill a high-dimensional pretrained

model on a carefully chosen representative complex alloy, and then use this single model to perform comparative studies across its subsystems. Such a “one-for-many” strategy reduces blind trial-and-error in exploring vast compositional spaces and can substantially improve the efficiency of both simulations and experiments by focusing follow-up work on the most promising compositional adjustments.

The predictive consistency between DPA2-All and DPA2-5E in cascade simulations originates from the fact that, for random atomic configurations, the difference in their energy and force errors is substantially smaller than the characteristic energy scale associated with defect production, i.e., the displacement threshold energy. As a result, both models yield very similar FPs statistics under irradiation, despite DPA2-5E having a

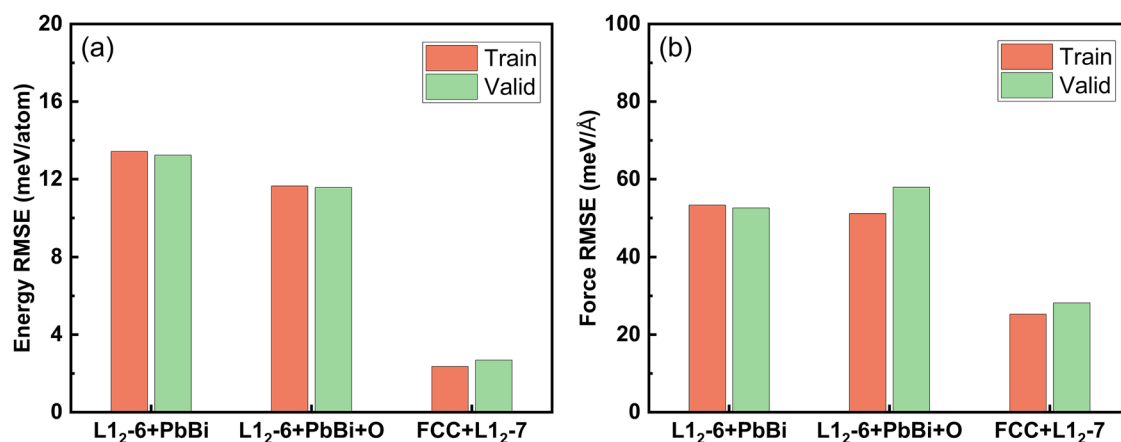


Fig. 6 | Energy and force prediction performance of MLIAPs constructed using pretrained LAM strategies for three representative complex systems. The “L1₂-6 + PbBi + O” system consists of a solid L1₂-6 structure (Ni, Co)₃(Al, Ti, Ta, Nb), a surrounding Pb-Bi liquid phase, and an oxide phase; The “FCC + L1₂-7” system

includes a random FCC matrix (Ni-Co-Fe-Cr-Al-Ti-Ta) and coherent L1₂ precipitates of (Ni, Co)₃(Fe, Cr, Al, Ti, Ta). Orange bars represent training RMSEs, and green bars represent validation RMSEs. **a** Energy prediction error. **b** Force prediction error.

somewhat larger force RMSE in certain out-of-domain environments. In line with this, DPA2-All, with its broader chemical and structural diversity, is naturally more suitable as a universal potential for exploring a wide range of ordering motifs and compositions beyond the Ta-Nb-W-Mo-V quinary, whereas DPA2-5E, constructed from a much smaller quinary-focused dataset, already reasonably captures the cascade and mechanical response of the target system and its subsystems with good fidelity at substantially reduced data and training cost. This illustrates a practical trade-off between universality and efficiency: a broadly trained model such as DPA2-All offers greater robustness for chemically diverse systems, while a targeted, fine-tuned variant like DPA2-5E provides an efficient and accurate tool for in-depth studies of a specific complex alloy and its closely related subsystems. A systematic comparison between explicitly short-range-ordered and fully random configurations, and an assessment of how chemical short-range order modifies irradiation response within this framework, remain important directions for future work.

Importantly, these findings are not limited to alloys with complex compositions. The same LAM-based DPA2 strategy can be extended to systems with complex phase structures or exposed to other harsh nuclear reactor conditions, such as corrosion and oxidation. To further evaluate its applicability, as shown in Fig. 6, we applied the strategy to ordered inter-metallic compounds—specifically, dual-phase FCC + L1₂ microstructures in the Ni-Co-Fe-Cr-Al-Ti-Ta system—and to a pure L1₂-phase alloy (Ni-Co-Al-Ti-Ta-Nb) exposed to corrosive liquid-Pb-Bi environments, with and without oxygen. Even under these challenging structural and environmental conditions, the LAM-based MLIAPs achieved high prediction accuracy with minimal additional training data, reinforcing the versatility and transferability of the strategy. These results expand the applicability of our method beyond compositional complexity to structural and environmental complexity, offering a unified, data-efficient strategy for modeling a wide spectrum of nuclear materials. This opens new avenues for investigating previously intractable problems—such as corrosion-irradiation synergy and environment-induced degradation—in alloys featuring both complex chemistries and microstructures. The training details are provided in Supplementary Materials S2.

Despite the strong transferability and wide applicability demonstrated in Fig. 3, we observe a consistent and mild underestimation of yield strength by the DPA2-5E model across all tested alloys—including the quinary Ta-Nb-W-Mo-V, quaternary Ta-Nb-W-Mo, and binary Ta-W systems (Fig. 5). This systematic deviation does not compromise the qualitative temperature-dependent trends or comparative performance assessments. Rather, it highlights an important nuance in extending MLIAPs to predict emergent mechanical properties. This systematic softening offers deeper insight into the

intrinsic limitations of pretrained MLIAPs: most pretraining datasets are dominated by near-equilibrium atomic arrangements, which biases the learned potential energy surface and results in underpredicted energies and forces under non-equilibrium conditions^{45,46}. Recent studies have shown that this softening can be efficiently mitigated by incorporating non-equilibrium data during fine-tuning⁴⁵. Our results further support this conclusion: after including non-equilibrium data from the quinary system in the DPA2-5E fine-tuning, the model’s yield strength predictions for the quinary alloy show minimal deviation (see Fig. 5a). However, as the composition shifts from quinary to quaternary and binary alloys (see Fig. 5b, c), the prediction errors progressively increase, indicating that the benefit of fine-tuning with quinary non-equilibrium data diminishes as the target system deviates from the fine-tuned composition, and the model’s performance reverts closer to its original pretrained capacity, thus amplifying the softening effect.

We further quantified how non-equilibrium fine-tuning alters the yield-strength errors of DPA2-5E relative to the full-data reference model DPA2-All (Supplementary Materials S4). For the target Ta-Nb-W-Mo-V alloy, the original DPA2-5E model—already fine-tuned with a moderate amount of quinary non-equilibrium data—shows a mean deviation of about 1 GPa over the explored temperature range, which is consistent with the results shown in Fig. 5a. After adding 1200 extra non-equilibrium quinary compression configurations, the mean deviation remains at approximately 1 GPa, indicating that no substantial improvement is achieved with the additional data. This suggests that, for this particular model and training setup, the benefit from incorporating quinary non-equilibrium data has already reached a plateau after the initial fine-tuning, and further increasing the quinary data volume yields negligible additional quantitative gain for the quinary yield-strength predictions. In contrast, representative quaternary alloys exhibit a much larger quantitative improvement, with mean deviations reduced from about 2.56–0.61 GPa, whereas binary alloys show only a modest change (from 4.18 to 3.79 GPa), and the systematic softening is not fully removed. These numbers provide a concrete, system-specific example of how targeted non-equilibrium fine-tuning can substantially benefit some subsystems while only partially correcting others. At the same time, we do not intend them to define a universal quantitative relation between data volume and prediction error: the effect of adding non-equilibrium data also depends on model capacity, optimization schedule, total training compute, and the diversity and relevance of configurations, consistent with recent scaling-law analyses⁴¹.

The observed transferability of the quinary-fine-tuned DPA2-5E model is rooted in how the pretrained LAM encodes local atomic environments, rather than being a purely mathematical artifact of RMSE aggregation. During pretraining, the model is exposed to a wide range of

elements and bonding motifs, and the learned type embeddings organize the periodic table into chemically meaningful patterns, effectively providing a global “map” of elemental interactions. Fine-tuning on Ta-Nb-W-Mo-V then locally refines this map in the region corresponding to the quinary alloy system, making the energy and force landscape in that subspace more accurate while retaining the global chemical trends. Our t-SNE analysis (Supplementary Materials S5) of local environment descriptors shows that quinary configurations occupy a distinct but overlapping region with their quaternary, ternary, and binary subsystems, and that prediction errors increase as these environments move further away from the quinary-dominated region. This behavior indicates that the model is not simply memorizing specific compositions but is leveraging shared elemental characteristics and bonding motifs across related alloys, with accuracy degrading systematically as local environments depart from those most strongly constrained during fine-tuning. In this sense, the good performance of DPA2-5E on many lower-component subsets reflects a physically grounded transfer of local environment knowledge from the quinary HEA to its subsystems, rather than a purely statistical coincidence.

For additional context, we also benchmarked the pretrained and fine-tuned DPA2 models against conventional empirical EAM/MEAM-type potentials on three ternary alloys in the Ta-Nb-W-Mo-V chemical space (W-Ta-V, W-Mo-Nb, and W-Ta-Mo), where such empirical IAPs are available^{47–49}. As shown in Supplementary Materials S6, both DPA2-All and DPA2-5E reduce the energy training errors from values on the order of several eV/atom for the empirical potentials to below ~10 meV/atom, and lower the force RMSE by roughly an order of magnitude. These results highlight that, even before any system-specific enrichment for phenomena such as complex defect structures, the pretrained-LAM-based MLIAPs provide a far more accurate description of atomic interactions in refractory alloys than currently available empirical potentials. In this sense, empirical EAM/MEAM models remain useful for very large-scale, low-cost exploratory simulations, but the DPA2-based MLIAPs offer a more reliable starting point whenever quantitative predictions of defect energetics, cascade response, or mechanical behavior in complex multicomponent systems are required.

We note several limitations of the present study and offer some practical guidance for using these models in future work. Our assessment of the pretrained-LAM framework is based on energy/force accuracy, cascade defect statistics, and temperature-dependent mechanical response in the Ta-Nb-W-Mo-V family (with additional tests on defect-related configurations), and our main conclusion—that a single quinary-fine-tuned model such as DPA2-5E can describe closely related subsystems with reasonable accuracy—should be understood within this scope. For applications that involve highly heterogeneous microstructures, strong elemental segregation, interfaces, or properties that are extremely sensitive to small energetic differences (e.g., diffusion barriers, phase competition, interfacial energetics), additional system- and property-specific fine-tuning will likely be necessary, and the more broadly trained DPA2-All model may remain the safer default choice. In this sense, the present work should be viewed as providing a hierarchy of options rather than a universal solution: DPA2-All as a general-purpose, broadly transferable potential, and DPA2-5E as a computationally efficient alternative for Ta-Nb-W-Mo-V and its subsystems when the target properties are similar to those explored here. We hope this clarification helps readers select the most appropriate modeling strategy for their own studies of complex multicomponent alloys.

Beyond the cascade and mechanical tests presented here, a key next step is to exploit the present DPA2-based MLIAPs for quantitative studies of point-defect diffusion and post-cascade microstructure evolution in Ta-Nb-W-Mo-V alloys. Our additional benchmarks on defect-related and transition-state configurations (Supplementary Materials S7) already show that the pretrained models achieve substantially higher accuracy in defect energetics and forces, providing a reliable foundation for such applications. Building on this, systematic simulations of diffusion, segregation, and long-time annealing—e.g., via large-scale MD coupled with KMC—represent a natural direction for future work to further assess and exploit the capabilities of pretrained-LAM-based MLIAPs in complex nuclear alloys.

Overall, the observed behavior of DPA2-5E suggests a practical and scalable modeling strategy for complex nuclear materials: the pretrained and fine-tuned model provides a robust baseline for rapid screening and initial property assessment across a wide range of chemically complex systems. When higher fidelity is needed for a specific nuclear alloy, targeted fine-tuning with a small amount of non-equilibrium data from the subsystem can substantially improve prediction accuracy. This hierarchical workflow enables efficient inverse design, rapid mechanical property evaluation, and cross-system comparison—especially crucial for complex nuclear materials, where the vast compositional space and scarcity of high-quality labeled data make traditional approaches infeasible.

Methods

Dataset and preprocessing

The dataset of Ta-Nb-W-Mo-V and all its equimolar subsystems (“All” dataset) comprises a diverse set of alloy combinations, ranging from unary (single element) to the full quinary (5-element) system^{42,44}. To evaluate the generalization and transferability of the LAM-based DPA2 approach and the scratch-training DPA1 method over different alloy systems and their subsystems, we constructed MLIAPs using only the quinary dataset and the “Dataset-All” dataset. The first approach generated MLIAPs⁴⁰, termed DPA2-5E (trained on DPA2 and the dataset on 5-element alloy system only), was obtained by fine-tuning a multi-task pretrained large model within the LAM-based DPA2 framework; the second approach generated MLIAPs, DPA1-5E (trained on DPA1 and the dataset on 5-element alloy system only), was trained from scratch using the DP-based attention method³³. For comparison, we also trained reference MLIAPs using the “Dataset-All” datasets: DPA2-All and DPA1-All.

During data preprocessing, in order to keep the composition and structure diversity, the training sets not only include the composition configurations from the unary to quinary alloy crystals, but also consider both the different vacancy and interstitial configurations, liquid and surface phases, and transition-state data. The detailed data preprocessing can be found in the Supplementary Materials S1. All data processing was performed using the *dpdata* package to guarantee that the final dataset format was compatible with both LAM-based DPA2 and standard DPA1 training, as illustrated in Fig. 2a.

MLIAPs training

Figure 2b outlines the training workflows for the LAM-based DPA2 approaches. For the LAM model, we adopted the **DeePMD-kit v3** code⁵⁰ to fine-tune based on a pretrained model *DPA2_medium_28_10M_rc0.pt* from the AIS Square website (www.aissquare.com/models/). Under a matched pretraining budget (batch size = 1; 1×10^6 steps) on identical datasets, DPA2 reported the lowest energy WARMSE and the second-lowest force WARMSE among GemNet-OC, Equiformer-V2, NequIP, Allegro, and MACE, suggesting favorable generalization in the low-resource regime⁴⁰. Note that DPA2, NequIP, and Allegro are conservative (forces = $\partial E/\partial R$), whereas GemNet-OC and Equiformer-V2 are not, which may influence comparative force-error rankings.

In the fine-tuning stage, the branch-head named *Domains_alloy* (which supports up to 53 metal types) is used to accommodate the multiple elements in our target systems. This produced a “teacher” model (denoted DPA2) with a substantial number of parameters. Specifically, we trained the DPA2 model for 1×10^6 steps using the *dpa2* type descriptor and the neural network architecture, which consists of two sub-networks: an embedding network of three layers (with 25, 50, 100 nodes, respectively) and a fitting network of three layers (240 nodes each). A uniform cutoff radius of 6.0 Å was applied to all elemental interactions in the embedding descriptor. All other training hyperparameters are provided in our publicly available training scripts and results repository (<https://github.com/JiangXiaoMingSan/LAM-DPA-Train-Script>).

To address the low computational efficiency of the fine-tuned teacher model, we applied a knowledge distillation procedure to obtain a simpler “student” model (Fig. 2b). In each distillation iteration, the teacher model

(DPA2) was used to label a set of configurations, and the student model was retrained on these to mimic the teacher. We used the DP-GEN 2.0 platform⁵¹ to orchestrate the iterative training process by the DPA1 framework³³, employing a lighter *se_atten_v2* descriptor while keeping the same fitting network structure as the teacher. Each distillation round involved 1×10^6 training steps. We monitored the prediction discrepancy between the teacher and student models on the validation configurations. If the student's error exceeded a predefined threshold, the corresponding configurations were added to the training set for the next iteration. This process was repeated until the student model reached the target accuracy, as determined by an adaptive tolerance scheme (with $\text{conv_tolerance} = 0.02 \text{ eV/\AA}$). To enhance the temperature-dependent transferability of the model, we further augmented the distillation process with MD sampling at seven temperatures: 300, 673, 973, 1273, 1573, 1873, and 2100 K. Approximately 4000 representative structures were extracted from each MD trajectory and incorporated as additional training data in successive distillation rounds. The resulting distilled student model retains the accuracy of the teacher model while significantly reducing computational cost. For the DPA1 model, the training process is more straightforward (Fig. 2c). We built the DPA1 model using the standard *se_atten_v2* descriptor and the same neural network architecture as above, and trained it on the dataset for 1×10^7 steps (since no pretraining is involved, a much longer training is needed).

Simulation methods

We carried out large-scale MD simulations using the LAMMPS code³⁷. Short-range nuclear repulsion was modeled by the Ziegler-Biersack-Littmark (ZBL) universal repulsive potential⁵² for interatomic separations below the chosen cutoff; the ML potential was smoothly joined to the ZBL potential in the 1.0–1.7 Å range using DeePMD-kit's embedded connection scheme, the detailed process is shown in the Supplementary Materials S3.

For the radiation damage (cascade) simulations, we used a cell of size $40 a_0 \times 40 a_0 \times 40 a_0$ (where a_0 is the lattice constant). Prior to introducing a cascade, the cell was equilibrated for 20 ps in the isothermal-isobaric ensemble (NPT) at 300 K. The cascade was then simulated by giving a PKA energy of 2 keV, 5 keV, and 10 keV. To avoid artificial channeling, the PKA was directed along the $\langle 111 \rangle$ crystallographic direction. During the cascade, the three outermost atomic layers of the cell were canonical ensemble (NVT) at 300 K, serving as a heat bath²⁸, while the interior evolved in the microcanonical ensemble (NVE). Each cascade was simulated for about 20–30 ps until defect counts stabilized.

For mechanical testing, we prepared elongated cells ($12 a_0 \times 12 a_0 \times 55 a_0$) with a single crystal to simulate uniaxial compression. Simulations were conducted at temperatures from 0 K up to 2100 K (in 300 K increments). At each temperature, the sample was first equilibrated for 50 ps in NPT (zero pressure) to establish the desired temperature and pressure conditions. Compression was then applied along the longitudinal z direction under NVT conditions at a high strain rate of $1.0 \times 10^9 \text{ s}^{-1}$ ^{53,54}. The transverse x and y directions were maintained at zero pressure during deformation. We continued the deformation for ~200 ps, long enough to reach significant plastic strain.

All atomic-scale visualization and analysis of defects were performed using the OVITO software package⁵⁵. Point defects were identified by the Wigner-Seitz cell method⁵⁶, where vacancies are counted by empty lattice sites and self-interstitials by occupying multiple atoms in a single lattice cell relative to an initial perfect reference.

Data availability

The datasets generated and analyzed in this study, along with the training scripts, example input files for the MLIP models, and all associated code for data processing, model training, and analysis, are publicly available in the GitHub repository at: <https://github.com/JiangXiaoMingSan/LAM-DPA-Train-Script>.

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Author contributions

Mingxuan Jiang and Biao Xu contributed equally to this work. Mingxuan Jiang carried out the data tests and drafted the initial manuscript. Biao Xu conceived and designed the research framework, coordinated the project, and supervised the overall study. Yixin Deng, Shihua Ma, and Fei Gao participated in data analysis and contributed to discussions and manuscript revisions. Ji-Jung Kai and Huiqiu Deng provided access to experimental facilities, funding, and essential resources. All authors discussed the results and contributed to the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Biao Xu, Ji-Jung Kai or Huiqiu Deng.

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